COMPREHENSIVE FRAUD & CUSTOMER ANALYTICS REPORT

EXECUTIVE SUMMARY

This report presents a complete analysis of transaction fraud patterns, customer segmentation, and spending behavior across six datasets. The analysis reveals critical insights into fraud distribution, temporal patterns, customer value distribution, and high-risk profiles.

Key Findings at a Glance:

- Total fraud exposure: **$89,249.84** across **1,000 incidents**
- All **546 tracked customers** are currently **inactive** (100% churn rate)
- **Grocery POS** is the highest-risk category ($20,347 — 22.8% of total fraud)
- Peak fraud activity occurs at **11:00 AM** (62 incidents)
- Only **1.8%** of customers are classified as "Medium Value" (10 out of 547)
- Top spender: **Benjamin Harris** at **$4,221.72**

SECTION 1: ACTIVE/INACTIVE CUSTOMER ANALYSIS

Dataset Overview

The Active_Inactive_Customers.csv contains **546 customer records** with status and inactivation dates spanning from January 2019 to June 2020.

Key Metrics

Metric	Value
Total Customers	546
Active Customers	0
Inactive Customers	546
Churn Rate	100.0%
Date Range	Jan 2019 –Jun 2020
Peak Inactivation Month	December 2019 (65 customers)

Analysis Insights

1. Complete Customer Churn: Every single tracked customer is inactive, indicating a severe retention crisis. This 100% churn rate is extraordinarily high and demands immediate investigation into root causes — whether due to account closures, fraud-driven exits, or data collection limitations.

2. Seasonal Inactivation Pattern: The monthly trend shows a significant spike in December 2019 (65 customers), suggesting potential holiday-season fraud impact or end-of-year account cleanup. The pattern shows gradual acceleration through 2019, peaking in Q4, then stabilizing in early 2020.

3. Yearly Distribution: 2019 accounts for **310 inactivations (56.8%)** versus **236 in 2020 (43.2%)**. The 2020 data is partial (through June), so annualized 2020 would likely exceed 2019.

4. Name Frequency Analysis: "Smith" is the most common surname (13 occurrences), followed by Davis and Jones (8 each). This demographic insight could help target retention campaigns.

SECTION 2: CUSTOMER SEGMENTATION ANALYSIS

Dataset Overview

Customer_Segmentation.csv contains **547 customer records** with transaction amounts and value classifications (Low/Medium Value).

Key Metrics

Metric	Value
Total Customers	547
Total Revenue	$89,249.84
Average Transaction	$163.16
Median Transaction	$90.90
Minimum Transaction	$1.03
Maximum Transaction	$4,221.72
Low Value Customers	537 (98.2%)
Medium Value Customers	10 (1.8%)

Analysis Insights

1. Extreme Value Concentration: The customer base is heavily skewed toward low-value transactions. With 98.2% classified as "Low Value," the business has minimal high-value customer penetration. The 10 medium-value customers generate disproportionate revenue relative to their count.

2. Revenue Distribution: Total revenue of $89,249.84 comes almost entirely from low-value customers ($70,686), while medium-value customers contribute $18,564. Despite being only 1.8% of the base, medium-value customers account for **20.8% of revenue**.

3. Transaction Amount Distribution: The histogram reveals a heavily right-skewed distribution. Most transactions cluster below $500, with a long tail extending to $4,221.72. The median ($90.90) is significantly lower than the mean ($163.16), confirming the presence of high-value outliers pulling the average upward.

4. Top Customer Profiles: Benjamin Harris ($4,221.72), Ashley Cruz ($3,208.68), and Lisa Mendez ($2,491.70) represent the top tier. These three customers alone account for **$9,922.10** — over 11% of total revenue.

SECTION 3: FRAUD ANALYSIS BY CATEGORY

Dataset Overview

fraud_count_category.csv contains **14 transaction categories** with fraud counts and total fraud amounts.

Key Metrics

Metric	Value
Total Fraud Amount	$89,249.84
Total Fraud Incidents	1,000
Average per Incident	$89.25
Highest Risk Category	Grocery POS ($20,347)
Most Frequent Category	Grocery POS (178 incidents)

Category Breakdown (Top 5)

Category	Count	Amount	Avg/Incident	% of Total
Grocery POS	178	$20,347	$114.31	22.8%
Shopping Net	114	$18,893	$165.73	21.2%
Gas Transport	149	$9,616	$64.49	10.8%
Shopping POS	148	$9,191	$62.10	10.3%
Misc POS	79	$7,987	$101.10	8.9%

Analysis Insights

1. Grocery POS Dominates Risk: Despite not having the highest average per incident ($114.31), Grocery POS leads in both total amount ($20,347) and incident count (178) due to high transaction volume. This makes it the #1 priority for fraud prevention.

2. Shopping Net — High-Value Threat: With an average of **$165.73 per incident**, Shopping Net has the second-highest per-incident cost among major categories. While fewer incidents than Grocery POS, each fraud event is more costly.

3. Travel — Highest Per-Incident Cost: At **$244.69 per incident**, Travel fraud is the most expensive per event, though with only 6 incidents. This suggests targeted, high-value fraud in travel bookings.

4. Low-Frequency Categories: Health Fitness ($26.96 avg), Kids Pets ($35.35), and Personal Care ($72.82) show lower per-incident costs but still contribute to overall exposure.

5. Count vs. Amount Scatter Plot: The scatter analysis reveals two distinct fraud profiles: high-frequency/low-amount (Grocery POS, Gas Transport) and low-frequency/high-amount (Travel, Shopping Net). Prevention strategies should differ for each profile.

SECTION 4: HOURLY FRAUD PATTERN ANALYSIS

Dataset Overview

Hourly_Frauds.csv contains **24 hourly records** showing fraud incident distribution across a full day.

Key Metrics

Metric	Value
Total Daily Incidents	1,000
Peak Hour	11:00 AM (62 incidents)
Lowest Hour	9:00 PM (26 incidents)
Average per Hour	41.7
Morning (6-11)	313 incidents
Afternoon (12-17)	190 incidents
Evening (18-23)	195 incidents
Night (0-5)	302 incidents

Analysis Insights

1. Morning Peak Risk: The 11:00 AM peak (62 incidents, 6.2% of daily total) suggests fraudsters exploit mid-morning transaction volumes when systems may be processing high volumes of legitimate transactions. This could indicate coordinated attacks during business hours.

2. Nighttime Vulnerability: Surprisingly, the Night period (0-5 AM) accounts for **302 incidents (30.2%)**, nearly matching the Morning period. This contradicts typical assumptions that fraud decreases overnight and may indicate automated/bot-driven fraud activity during low-staffing hours.

3. Afternoon Lull: The Afternoon period (12-17) shows the lowest activity at 190 incidents. This could represent a natural dip or effective afternoon monitoring.

4. Above-Average Hours: 11 hours exceed the daily average of 41.7 incidents. These concentrated risk windows should receive enhanced monitoring and real-time alerts.

5. Heatmap Visualization: The color-coded hourly chart clearly shows the concentration of risk in early morning hours (0-3 AM) and late morning (10-11 AM), with a secondary cluster around midnight.

SECTION 5: MASSIVE VELOCITY FRAUD ANALYSIS

Dataset Overview

MassiveVelocityFrauds.csv contains **8 high-velocity fraud cases** with customer details and amounts.

Key Metrics

Metric	Value
Total Cases	8
Total Amount	$4,337.25
Average per Case	$542.16

Table 6 – continued

Metric	Value
Median per Case	$603.77
Maximum	$1,007.47 (Nathan Byrd)
Minimum	$8.37 (Tammy Maldonado)
All Cases	Velocity = 1

Case Detail Table

Customer	Amount	Above Avg?
Nathan Byrd	$1,007.47	YES
Paula Estrada	$991.51	YES
Robert James	$882.64	YES
Jennifer Vega	$792.10	YES
Jeremy Chavez	$315.44	NO
Jeffrey Lewis	$290.25	NO
Denise Barnett	$49.47	NO
Tammy Maldonado	$8.37	NO

Analysis Insights

1. Bimodal Distribution: The cases split into two clear groups: 4 high-value cases ($792-$1,007) and 4 low-value cases ($8-$315). This bimodal pattern suggests two distinct fraud mechanisms — possibly manual high-value attacks versus automated low-value testing.

2. Top 4 Cases Dominate: The four cases above average ($542.16) account for **$3,673.72** (84.7% of total velocity fraud). These should be prioritized for investigation and recovery.

3. Velocity = 1 Across All Cases: Despite the "massive velocity" label, all cases show velocity of 1, suggesting this may represent the first detected incident in a sequence rather than rapid-fire transactions. This could indicate early-stage detection.

4. Customer Overlap: Jennifer Vega, Robert James, and Nathan Byrd appear in both velocity fraud and top spender lists, suggesting these high-value customers are also high-risk targets.

SECTION 6: TOP SPENDERS ANALYSIS

Dataset Overview

Top10Spenders.csv contains **55 high-value customers** with cities and spend amounts.

Key Metrics

Metric	Value
Total Tracked	55
Total Spend	$43,040.25

Table 8 – continued

Metric	Value
Average Spend	$782.55
Median Spend	$580.97
Minimum	$340.95
Maximum	$4,221.72
Unique Cities	55 (all distinct)

Top 10 Spenders

Rank	Customer	City	Amount
1	Benjamin Harris	Chester	$4,221.72
2	Ashley Cruz	Saint Amant	$3,208.68
3	Lisa Mendez	Lahoma	$2,491.70
4	Matthew Lambert	New Holstein	$1,993.99
5	Bianca Scott	Atglen	$1,305.09
6	Paula Estrada	Spencer	$1,113.99
7	Nathan Byrd	Red Cliff	$1,105.50
8	Kimberly Rice	Tyler	$1,074.79
9	Robert James	Tamaroa	$1,034.13
10	Jessica Ward	Phoenix	$1,014.47

Analysis Insights

1. Extreme Concentration: The top 3 customers (Benjamin Harris, Ashley Cruz, Lisa Mendez) account for **$9,922.10** — nearly 23% of total top-spender revenue. This heavy concentration creates significant revenue risk if any of these customers churn or become fraud victims.

2. Geographic Dispersion: All 55 customers are from unique cities, indicating a broad geographic spread with no single market dominance. This suggests a national or multi-regional customer base.

3. Spend Distribution: The histogram shows most customers cluster between $500-$1,000, with a sharp drop-off above $2,000. Only 4 customers exceed $2,000, making them exceptionally valuable and warranting VIP treatment.

4. Cumulative Curve: The top 10 customers generate approximately $18,564 (43% of total top-spender revenue). This Pareto distribution confirms that a small percentage of customers drive the majority of value.

5. Cross-Dataset Overlap: Several top spenders also appear in fraud datasets (Benjamin Harris, Ashley Cruz, Lisa Mendez in segmentation; Nathan Byrd, Paula Estrada, Robert James, Jennifer Vega in velocity frauds). This overlap between high-value and high-risk customers is critical for targeted protection.

CROSS-DATASET INSIGHTS & RECOMMENDATIONS

Critical Risk Intersections

1. **High-Value + High-Risk:** 4 customers appear in both top spender and fraud datasets
2. **100% Inactive Rate:** Complete customer churn suggests systemic issues beyond fraud
3. **Morning Peak + Grocery POS:** The 11 AM peak aligns with Grocery POS peak activity

Strategic Recommendations

1. **Immediate:** Investigate 100% customer inactivation — this is not normal
2. **Fraud Prevention:** Deploy enhanced monitoring for Grocery POS between 10 AM - 12 PM
3. **Customer Retention:** Target the 10 medium-value customers for retention programs
4. **Night Monitoring:** Increase automated monitoring for 0-5 AM period
5. **VIP Protection:** Implement dedicated fraud protection for top 10 spenders